Panskura Banamali College (Autonomous)

B.Sc.(H) 3rd Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC5P**

Paper Name : DBMS Lab

F.M. : 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2021UG009967, paper_code : CC5P , then file name will be :: 2021UG009967- CC5P.pdf for question 'A' and 2021UG009967- CC5P.mp3 for question 'B']

A. Answer any one question **1×12=12**

1. Consider the following relations

Student(snum, sname,major,level,age)

Class(name,meets,at,room,fid)

Enrolled(snum,fname,deptid)

Write the following queries in SQL

a) Find the names of all juniors(level=JR) who are enrolled in a class taught by M. Kumar

b) Find the names of all students who are enrolled in two classes that meet at the same time.

c) Find the names of the students who are enrolled in the maximum number of classes.

d) Find the names of students who are not enrolled in any class.

2. Consider the following relation

Sailors(sid,sname,ratting,age)

Boats(bid,bname,color)

Reserves(sid,bid,day)

a) Find the names of sailors who have reserved boat number 103.

b) Find the sides of all sailors who have reserved red boats but not green boats.

c) Find the name of sailors who are older than the oldest sailors with a rating of 10.

d) For each red boat find the number of reservation for this boat.

3. Consider the following relational database

Doctor(Dno.,Dname,Addr,City)

Hospital(Hno.,Name,Street,City)

Doc_Hosp(Dno.,Hno.,Date)

a) Find out all doctors who have visited hospitals in same city in which they live.

b) Find which hospital “Dr. Joshi” has visited ?

c) Count number of doctors visited to “Shree Clinic” on 1st March 2021.

d) Find out how many times “Dr. Joshi” has visited “Shree Clinic”.

4. Consider the following relations

Account(br_name,acc_no,bal)

Depositor(Cu_name,acc_no)

Customer(cu_name,cu_street,city)

a) Find the average balance for each customer who live in “DElhi” & has atleast 3 accounts.

b) Find the number of depositors for each branch.

c) Find list of account holders name.

d) Find the depositors name.

5. Consider the following relations

Project(pno,pname,chief_architect)

Employee(eno,ename)

Assigned_to(pno,eno)

a) Get the details of employee working on project on “RDBMS”

b) Find the employee numbers of employees who do not work on project “Robotics”

c) Get details of employee working on all database projects.

d) Get employee number of employee who work on at least one project that employee ‘2000’ works on.

6. Consider the following database :

hotel (h_no, h_name, h_address)

room (r_no, h_no, type, charge)

booking (h_no, g no, date_from, date_to, r_no)

guest (g_no, g_name, g_address)

(a) List all single rooms with a charge below rupees 2000.

(b) List all unoccupled room for the 'Grand hotel'.

(c) List the guest details currently staying in 'Seahawk' hotel.

(d) List the details of all guests staying at the 'Grand hotel from 26th June 2013.

(e) Find the name of the hotel which offers a room with cheapest charge.

7. Consider the following database :

Project (Project#, P_name, Chief-Architect)

Employee (Emp#. E_name)

(a) Get details of all employees in descending order.

(b) Get Emp# of employees working on project no 353.

(c) Get details of employees working on more than two projects.

(d) Get details of employees working on a project named 'Database'.

(e) List the names of the employees who work under the Chief Architect named S. Roy.

8. Salesperson(SSN,Name,State_year,Dept_no)

Trip(SSN,From_city,To_city,Departure_date,Return_date,Trip_Id)

Expense(Trip_Id,Account,Amount)

Give the details for trips that exceeded Rs. 5000 in expenses.

a) Print the SSN at salesman who took trips to 'Bangkok'

b) Print the total trip expenses incurred by salesman with SSN='786-89-7980'

c) List the names of all salesperson who have taken at least one trip of 10 days long.

d) Find the details of the trip in which highest expense occurred.

B. Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is SQL?
2. Difference between char & varchar .
3. What is domain .
4. Define primary key .
5. What is aggregation?
6. What is Denormalization ?
7. What is an Entity ?
8. What is weak entity set ?
9. What are the three levels of data abstraction ?
10. What is the difference between shared lock and exclusive lock ?

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